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Metalloradical intermediates in electrocatalytic reduction of CO₂ to CO: Mn versus Re bis-N-heterocyclic carbene pincers†

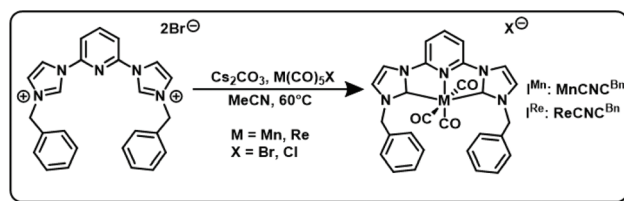
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This work examines the relative reactivities of Re^I and Mn^I tricarbonyl pyridine-2,6-bis-N-heterocyclic carbene pincers M(CO)₃CNC^{Bn}X (M = Re, Mn and X = Cl and Br) towards catalysis for the electrochemical conversion of CO₂ to CO. Unlike prior well-studied group VII catalysts, Mn(CO)₃CNC^{Bn}X is extraordinarily active, while the new Re(CO)₃CNC^{Bn}X complex surprisingly does not exhibit catalytic response. DFT calculations shed light on this puzzling behavior and show that the redox-active pyridine-2,6-bis-N-heterocyclic carbene ligand facilitates the reduction of the ground-state complexes; however, the extent of electronic delocalization in the reduced intermediates differs in the degree of metalloradical character. The highly-active Mn(CO)₃CNC^{Bn}X complex proceeds through an intermediate with nucleophilic metalloradical character in which 66% of the unpaired electron spin resides on Mn. In contrast, Re(CO)₃CNC^{Bn}X reduction proceeds through an intermediate with less metalloradical character in which only 38% of the unpaired spin is localized on Re with the remainder delocalized over the ligand. The energetic penalty of the electron delocalization of an electron on the ligand affects the M–CO bond strengths and related kinetic barriers. We discuss these observations in the context of turnover-enabling effects in CO₂ reductions mediated by group VII NHC pincer molecular electrocatalysts.

With an increased focus on catalytic reactions that convert abundant carbon sources such as carbon dioxide into useful fuels and chemicals, the discovery of new and scalable processes is at the forefront of research goals in the field of energy science. In order to achieve multielectron catalysis, organic ligands have been proposed to help these catalyses by acting as reservoirs of additional electrons for metals that would otherwise not be able to accommodate more than a

single 1e[−] process at a time, which is typical of earth-abundant first-row metals.^{1,2} Over the past decade, such redox-active ligands have been shown to enable new catalytic reactions and mechanisms that had not been previously accessible. As such, they continue to present an active direction for the development of multi-electron catalytic systems.^{3–7} Electrode-driven catalysis also benefits from its ease of scale-up as well as the opportunity to take advantage of light-driven technology where potential may be generated with light.⁸

Unfortunately, the use of a redox-active ligand as an electron reservoir does not always lead to productive catalysis, including the case of CO₂ electrocatalytic reductions. For example, Sieh⁹ and coworkers observed irreversible CO₂ binding to the ligand where the ligand-centered reduction on a molybdenum monimine pyridine complex led to a ligand–CO₂ addition product. This type of observation raises questions about the nature of the orbitals available for electrons to populate and how they enable or hinder catalytic processes. In this work, we compare two group VII pincer complexes: a tridentate pyridine-2,6-bis-N-heterocyclic carbene ligated to Mn (I^{Mn}) and to Re to form the novel complex I^{Re} (Scheme 1). Density functional theory (DFT) shows that the first reduction of both complexes is facilitated by a ligand orbital, thus highlighting the redox-active nature of this pincer ligand. However, once the ligand is reduced, the two analogous organometallic species exhibit different behaviors: the Mn pincer is electroactive towards CO₂ and shows good turnover, while the Re pincer does not. Given these observations, we herein explore the


Scheme 1 Synthesis of I^{Mn} and I^{Re}.

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specific nature of the key orbital necessary for productive CO₂ reduction catalysis *via* the proposed intermediates. This comes in contrast to the work of Kubiak and coworkers, where the Re and Mn analogues of the M(bipy)(CO)₃X (Mn, Br and Re, Cl respectively) have both been shown active for CO₂ reduction electrocatalysis.^{10–12}

Manganese, the first-row transition metal analogue of rhenium, has several advantages as a catalytic metal. Mn is the third most common transition metal in the Earth's crust,¹³ and is $\sim 10^6$ times more abundant than Re. As a first-row metal, Mn prefers 1e[−] redox steps and is often found in paramagnetic states. In contrast, Re, as a third-row transition metal, prefers 2e[−] redox steps and paired spins. The present work builds upon the wealth of knowledge of the comparative reactivities of Re(CO)₃bpyCl (bpy = bipyridine) and Mn(CO)₃bpyBr.^{14–19}

A recent report on the activity of Mn(CO)₃CNC^{Bn}Br²⁰ raises the possibility of catalytic conversion of CO₂ at electrodes enabled by electron-donating N-heterocyclic carbene (CNC) pincer ligands, an uncommon coordination environment in electrocatalysis for these types of transformations. Richeson and coworkers have reported a Mn pincer with phosphine ligation that is also able to turn over under CO₂ in the absence of protons.²¹ Given the promising activity exhibited by this compound and the group VI literature precedent, we chose to investigate the comparative catalytic response of its Re analogue. For this purpose, two metal complexes were synthesized; a first-row complex, Mn(CO)₃CNC^{Bn}Br²⁰ **I^{Mn}** and a new third-row complex, Re(CO)₃CNC^{Bn}Cl **I^{Re}**. The ligand in both of these complexes is a pyridine-2,6-bis-N-heterocyclic carbene with a delocalized π -system as shown in Scheme 1.²⁰

The pincer ligand 1,1'-(2,6-pyridinediyl)bis(3-benzylimidazolium) dibromide and the MnCNC^{Bn} complex **I^{Mn}** were prepared as previously reported.²⁰ Synthesis of the metallated Re complex **I^{Re}** was achieved by heating the ligand, Re(CO)₅Cl, and Cs₂CO₃ in acetonitrile for 16 h at 60 °C (Scheme 1) inside a nitrogen-filled glovebox. After cooling, the crude mixture was filtered, the solvent removed under vacuum, and the resulting red solid dissolved in a minimal amount of dichloromethane/methanol of 94 : 6 v : v and pushed through a neutral alumina column. After removal of solvent *in vacuo*, **I^{Re}** was isolated as a red solid in 64% yield.

The cyclic voltammogram (CV) of **I^{Re}** (Fig. 1, black) exhibits two electrochemical reduction responses at 2.03 V and 2.55 V. While the first reduction wave appears irreversible, return waves emerge at higher scan rates, indicating a coupled chemical process. Calculation of the numbers of electrons associated with these redox waves were performed by the method of Donadt and coworkers and resulted in a 1e[−] reduction per wave per Re center (ESI S1†).²²

Surprisingly, in the presence of added CO₂, **I^{Re}** exhibited almost no observable electrocatalytic reactivity. The CV currents observed upon degassing with Ar increase slightly upon addition of CO₂ (Fig. 1 red) and a proton source (Fig. 1, blue). The extent of this increase indicates that **I^{Re}** does not act as a catalyst for the desired conversion of carbon dioxide to chemi-

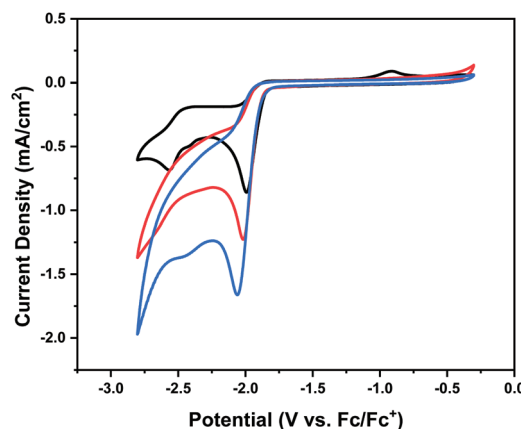


Fig. 1 Overlay of cyclic voltammograms collected at 100 mV s^{−1} of 2.5 mM **I^{Re}** under argon (black), CO₂ (red) and CO₂ with 5% TFE (blue).

cal fuels. The increase in current can be attributed to an increase in concentration of the sample upon degassing. Upon electrolysis at −2.5 V vs. Fc/Fc⁺ of a 2 mM solution of **I^{Re}** under both Ar and CO₂ with trifluoroethanol as proton source (5%), no quantifiable amount of CO is observed.

The reductive features of **I^{Re}** are almost identical to those of **I^{Mn}**, as shown in Fig. 2a with reduction potentials of −2.0 V vs. Fc/Fc⁺ for **I^{Re}** and −2.05 V for **I^{Mn}**, respectively. Re bipyridine carbonyl complexes are usually more difficult to reduce,¹⁷ therefore the nearly-identical first reduction potentials of the **I^{Re}** and **I^{Mn}** complexes suggest that the reduction may be ligand centered.

DFT calculations predict the first-reduction potentials at −2.23 V and −2.07 V for **I^{Mn}** and **I^{Re}**, respectively, and SOMOs with unpaired electronic spin density on the ligand framework. However, in the presence of CO₂, **I^{Mn}** exhibits an increased current response at the 1e[−] reduction wave at −2.05 V (Fig. 2b). In the presence of added protons, the current increase observed for Mn differs from the observed behavior of Re. To understand the observation that **I^{Re}** is unable to turn over we applied a DFT analysis. To qualitatively assess the catalysts' reactivity towards CO₂ reduction, we analyzed the computed electronic structures of the Mn- and Re-based starting materials and some putative preliminary intermediates formed upon reduction. The starting materials lack open reactive sites, but our calculations predict that CO release is favored upon reduction, thus providing at least one open reactive site on the metal center. In this investigation, the initial binding of a CO₂ molecule can be viewed as a reaction between a nucleophile and an electrophile. This analysis highlights several similarities as well as differences between the Mn- and Re-based complexes, for which we mainly focus on the factors that determine the reactivity of these complexes.

In our initial comparison of **I^{Mn}** and **I^{Re}** (Scheme 2), we found that the electron that adds to **I^{Mn}** and **I^{Re}** is not localized on the metal center, but instead delocalizes across the π -space of the aromatic ligand. Following reduction, the intermediates **III** are formed upon release of CO through the population of

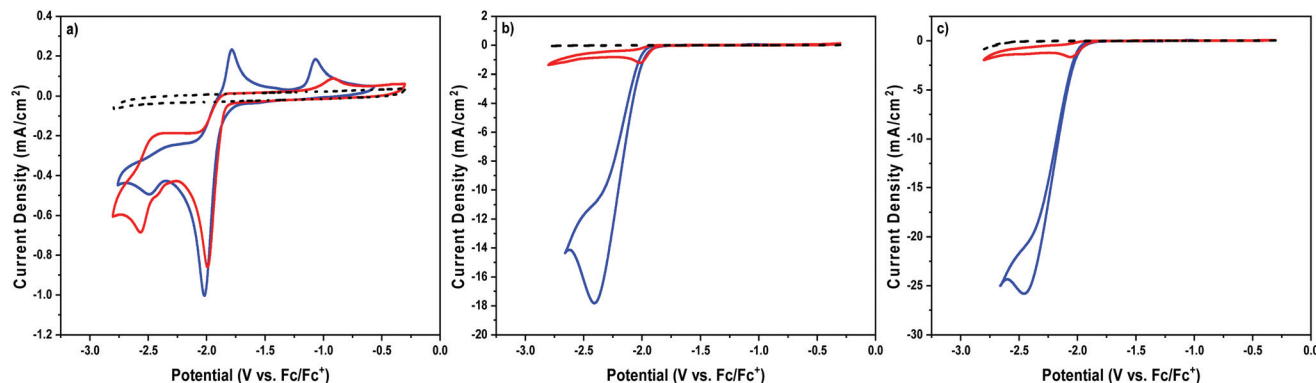
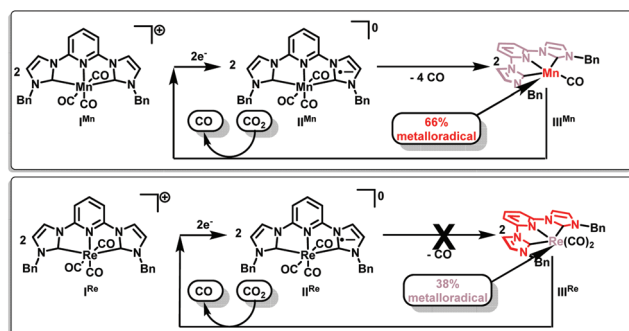


Fig. 2 Cyclic voltammograms of I^{Mn} and I^{Re} at 100 mV s^{-1} are shown in blue, and red, respectively. The black dashed line indicates the background control. Experiments were conducted in 5 mL acetonitrile with 0.1 M tetra-*N*-butylammonium hexafluorophosphate. Voltages are referenced against ferrocene as an external standard. (a) A comparison of the Re and Mn complexes under argon. (b) I^{Mn} and I^{Re} complexes with the addition of CO_2 in the absence of protons. Panel (c) I^{Mn} and I^{Re} after the addition of 5% TFE under CO_2 .



Scheme 2 Reduction of I^{Re} and I^{Mn} complexes.

an anti-bonding σ^* orbital. A spin population density analysis of III^{Mn} ($n_{\text{CO}} = 2$) and III^{Re} ($n_{\text{CO}} = 2$) shows that their different distributions of electron density over the metal and ligand leads to differences in the metalloradical character of the complexes where a larger metalloradical character correlates to a more reactive metal complex because the metal is the active center in CO_2 reduction.²³ In the case of III^{Mn} , we calculated that 66% of the spin density resides on the metal, with the remaining 34% delocalized over the ligand, as shown in Fig. 3a. For III^{Re} , the metalloradical character is attenuated to only 38% where now 62% of the spin population resides on the ligand (Fig. 3b). This is consistent with prior studies that showed that spin delocalization over the ligand is larger for

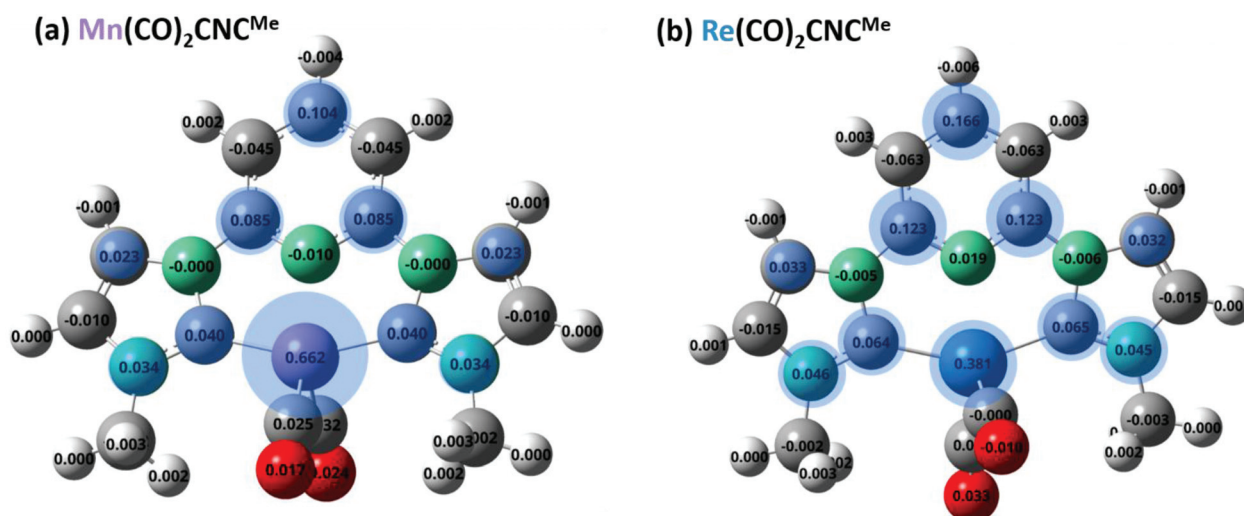


Fig. 3 Natural bond orbital spin population densities of (a) III^{Mn} ($n_{\text{CO}} = 2$) and (b) III^{Re} ($n_{\text{CO}} = 2$). Larger positive numbers indicate sites with the highest probabilities at which an unpaired electron resides. These probabilities are also illustrated with transparent blue spheres where a larger sphere indicates higher radical character, e.g., as indicated by the larger sphere on Mn than on Re. Details of the computational methods are provided in the ESI.†

second and third row transition metals relative to first row transition metals.²⁴ With III^{Re} being catalytically inactive, we tentatively attributed the lack of productive turnover of the III^{Re} complex to it not being sufficiently nucleophilic. An alternative description of the higher reactivity of III^{Mn} relative to III^{Re} can be illustrated by examining the differences in the energies of the valence s- and d-orbitals of Mn and Re as well as the spatial overlap between these orbitals and the p-orbitals of the CNC ligand. The distance between the Mn center and the CNC ligand is shorter than the Re–CNC distance (N–Mn = 2.02 Å, N–Re = 2.16 Å, Fig. S5.1 and S5.2†) because Mn is considerably smaller than Re. Furthermore, the 3d and 4s orbitals of Mn are closer in energy to the 2p orbitals of the CNC ligand compared to the energies of Re's 5d and 6s orbitals. These factors allow better mixing of atomic orbitals in the case of Mn, and as a result, greater molecular orbital splitting. As such, upon reduction, the added electron occupies an orbital that lies higher in energy in the Mn complex relative to its Re counterpart. Because the atomic orbitals of Re mix less extensively with the CNC ligand than Mn's atomic orbitals, more molecular orbitals of ReCNC – including LUMO/SOMO – lie near the Fermi level due to smaller orbital splitting, thus explaining the higher degree of electron delocalization in ReCNC relative to MnCNC. The localization of electrons on Mn (on a higher energy orbital) further validates its stronger nucleophilic character.

While the degree of metalloradical character provides a rationale for the drastic differences in reactivity between these complexes, our calculations also show that a unique intermediate exists which is accessible to the Mn complex but not to the Re complex. Upon a single $1e^-$ reduction of **I** to **II**, both the Mn and Re complexes can release one CO ligand. Moreover, our calculations reveal that the release of two CO groups (instead of a single CO) is uphill by 5 kcal mol^{−1} for the Mn complex, whereas the barrier for the release of two COs from the Re analog proceeds through a barrier of over 50 kcal mol^{−1}. This phenomenon may be explained by the higher degree of electron delocalization in the Re complex and the corresponding large increase in kinetic energy and Coulomb repulsion to localize an electron from the ligand to populate the Re–CO anti-bonding σ^* orbital, leading to the unfavorable release of the second CO. These results predict that Mn electrochemically accesses a unique square planar intermediate (III^{Mn} , $n = 1$; MnCNC–CO) that Re is less likely to attain and that participates in distinctive chemistries due its planar geometry. Its reactivity is the subject of a recent report.²⁵

In conclusion, the present study examined the differences in electronic structure between I^{Mn} and I^{Re} along with those of potential intermediates associated with their reduction. Our results suggest that the localization of unpaired electron density dictates whether the complexes will be catalytically competent through the nucleophilic character of the key metalloradical intermediate. While redox-active ligands enable reductions of metal complexes at lower overpotentials and can confer “nobility” to non-noble metals,²³ they may also divert

catalysis towards non-productive pathways. Our study puts forward the hypothesis that without the formation of a strong metalloradical, the interaction with the incoming CO₂ does not lead to productive turnover in the case of group VII organometallics such as I^{Mn} and I^{Re} .

Conflicts of interest

The authors declare no competing financial interests.

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